FILTER DESIGN TEST CASE

Coupled Resonator Filter - GPS L1 (1575 MHz) for 4G LTE Carrier Aggregation
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P2 - High

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting GPS L1 (1575 MHz) for 4G LTE Carrier Aggregation. The filter is designed on Langasite substrate with a target center frequency of 2362.4 MHz and bandwidth of 101.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: GPS L1 (1575 MHz)
- Application: 4G LTE Carrier Aggregation
- Substrate: Langasite
- Package: CSP (Chip Scale Package)
- Design Tool: Synopsys CustomSim
- Design Center: Singapore RF Lab

This specification applies to all variants of the Coupled Resonator Filter design targeting GPS L1 (1575 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 2362.4 MHz
Bandwidth (3 dB): 101.0 MHz
Insertion Loss Target: <= 1.2 dB
Return Loss Target: >= 20.0 dB
Out-of-Band Rejection: >= 43.0 dB
Isolation (Duplexer): >= 25.0 dB
Q Factor: 4215
Temperature Coefficient: -24.0 ppm/°C
Die Size: 1.6 x 1.3 mm
Wafer Size: 8-inch
Number of Resonators: 7
Electrode Material: Mo/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Synopsys CustomSim and load the Coupled Resonator Filter template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 2362.4 MHz, BW 101.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 7087 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 152 MHz offset >= 43.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 7.1 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 95%

6. TEST RESULTS

Measurement Date: 2024-11-02
Devices Tested: 51
Insertion Loss (meas): 0.86 dB
Return Loss (meas): 21.7 dB
Rejection (meas): 46.2 dB
Isolation (meas): 29.7 dB
Q Factor (meas): 4215
Group Delay Variation: 2.2 ns
Power Handling: +29 dBm
Die Yield: 80.6%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The Coupled Resonator Filter design demonstrated below-target performance for GPS L1 (1575 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Group delay flatness improved compared to previous revision.
e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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